

1. Team Details

Team name: PLACEHOLDER_TEAM [[FILL IN BEFORE SUBMISSION]]

Members (name -- role):

[MEMBER 1] -- modelling

[MEMBER 2] -- data & augmentation

[MEMBER 3] -- inference optimization

[MEMBER 4] -- evaluation & docs

College: [COLLEGE NAME]

Contact: [EMAIL] / [PHONE]

2. Problem Statement Addressed

AI-Based Restoration of Degraded Images for Semiconductor Inspection (KLA PS01). A single lost pixel or noisy region in an inspection image can hide a real defect and cost a die -- restoration quality is a correctness problem, not a cosmetic one.

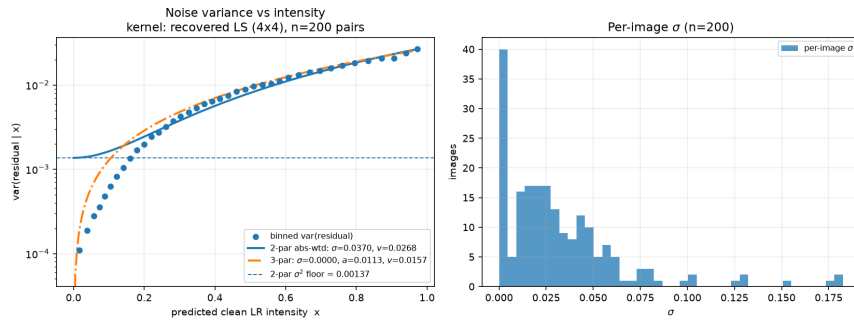
Three degradation mechanisms, applied in combination:

1. Speckle noise (signal-dependent, multiplicative-like variance term)
2. Additive/shot noise (signal-dependent linear term)
3. Downsampling (x2 decimation, GT/LR pair)

The pipeline must invert all three jointly, in whatever order they were applied, and generalise to content the training set never showed it.

3. Idea Description

The released dataset is 3200 training pairs and 400 test inputs of grayscale natural photographs, not semiconductor imagery. We treat it as a proxy: the degradation — $\times 2$ decimation plus signal-dependent noise — is what transfers to inspection imagery, so we characterised the degradation empirically and optimised for degradation robustness rather than fitting content-specific priors.



Measured (D1/D12/D2): downsample kernel is a recovered 4x4 sharpening kernel (bicubic antialias-off is within $1.22e-05$ of optimal); noise is applied AFTER downsampling, signal-dependent, no additive Gaussian floor (D2, residual autocorrelation ~ 0).

Core concept: one-step blind joint restoration, all compute at LR resolution, $\times 2$ PixelShuffle head -- no cascade, no GAN.

4. Proposed Solution

Pipeline: load .npy -> group by shape -> batch -> bf16 forward (channels_last) -> clip [0,1] -> save

Architecture: NAFSR -- NAFNet-style blocks (SimpleGate, SCA channel attention, LayerNorm) + x2 PixelShuffle head, width 48, 16 blocks

Loss: Charbonnier (fidelity) + SSIM (structure) + FFT (frequency) -- balanced, no adversarial term (no hallucination risk for inspection use)

Augmentation: dihedral flips/rotations, CutBlur, on-the-fly synthetic re-degradation with randomised order and noise levels

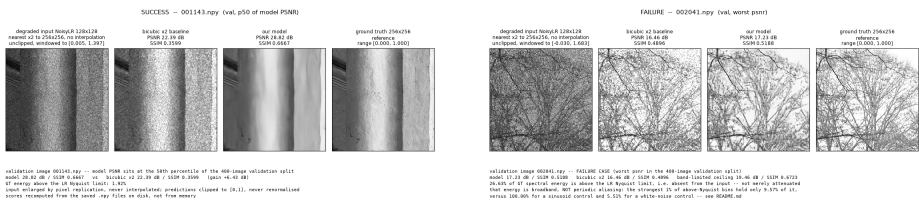
5. Innovation & Uniqueness

- (a) Empirical degradation forensics driving a matched synthetic-pair generator -- kernel and noise parameters measured from the data, not assumed from the spec.
- (b) Balanced fidelity + structure + frequency loss with an explicit no-GAN / no-hallucination decision -- justified by inspection semantics: a plausible but wrong structure is worse than a blurry correct one when a die's fate depends on it.
- (c) Throughput-engineered inference path: grouped batching by shape, bf16 + channels_last, threaded I/O, memory-aware OOM recovery (automatic batch halving, CPU-bicubic floor) -- measured, not assumed.

6. Results

Method	PSNR dB	SSIM	LPIPS	n
Bicubic x2 (raw NoisyLR)	23.6524 +/- 3.0236	0.54775 +/- 0.19197	0.41206 +/- 0.15407	400
Median 3x3 -> bicubic x2	25.5057 +/- 3.8785	0.61317 +/- 0.17232	0.40870 +/- 0.15866	400
Non-local means -> bicubic x2	26.2722 +/- 4.3037	0.65152 +/- 0.19523	0.42586 +/- 0.18627	400
U-Net baseline	28.8808 +/- 4.5328	0.78273 +/- 0.14245	0.26525 +/- 0.14878	400
Final model	28.7865 +/- 4.5329	0.78287 +/- 0.14169	0.25324 +/- 0.13193	400

Proxy-OOD (procedural geometric content, n=40): PSNR 27.3177 +/- 3.4696 / SSIM 0.96493 +/- 0.01167 / LPIPS 0.03797 +/- 0.01780 -- synthetic, not semiconductor imagery; degradation model unchanged, not refit.



Success (median case) | Honest failure: worst-PSNR case,
band-limited oracle ceiling caveat applies

7. Technology & Feasibility

PyTorch 2.11.0+cu128, CUDA 12.8, trained on NVIDIA RTX 4060 Laptop GPU (8 GB)

Model: NAFSR, 388,225 params, checkpoint 3.14 MiB, trained 20,000 iterations

Inference throughput (128->256, NVIDIA GeForce RTX 4060 Laptop GPU (compute capability 8.9, driver 610.47), bf16, batch 32): 8.3 img/s, fixed startup 30.6% of wall-clock at N=400 -- externally timed, process start to finish

Inference throughput (256->512, NVIDIA GeForce RTX 4060 Laptop GPU (compute capability 8.9, driver 610.47), bf16, batch 32): 11.9 img/s, fixed startup 34.3% of wall-clock at N=400

No H100 number is reported unless explicitly labelled as a projection with its formula shown -- KLA scores on an H100, we trained/measured on an RTX 4060 and (if landed) an A100 cloud run, and never conflate the two.

8. GitHub & Video Link

Repository: <https://github.com/sahithsundarw/semicon-kla-image-restoration>
(public, verified in a logged-out window)

Demo video: [[OPTIONAL, ≤ 5 min, link here if recorded]]

9. References

- Kumar, T. et al. (2024). Image Data Augmentation Approaches: A Comprehensive Survey and Future Directions. IEEE Access, 12.
- Zhai, L. et al. (2023). A Comprehensive Review of Deep Learning-Based Real-World Image Restoration. IEEE Access, 11, 21049-21067.
- Terven, J. et al. (2025). A Comprehensive Survey of Loss Functions and Metrics in Deep Learning. Artificial Intelligence Review, 58, 195.
- Monga, V. et al. (2021). Algorithm Unrolling: Interpretable, Efficient Deep Learning for Signal and Image Processing. IEEE SPM, 38(2), 18-44.
- Chen, L. et al. (2022). Simple Baselines for Image Restoration (NAFNet). ECCV.
- Yoo, J. et al. (2020). Rethinking Data Augmentation for Image Super-Resolution (CutBlur). CVPR.
- Wang, Z. et al. (2004). Image Quality Assessment: SSIM. IEEE TIP.
- Zhang, R. et al. (2018). The Unreasonable Effectiveness of Deep Features as a Perceptual Metric (LPIPS). CVPR.
- Shi, W. et al. (2016). Real-Time Single Image and Video SR Using an Efficient Sub-Pixel CNN (PixelShuffle). CVPR.